

with well

- ✓ 1. While highlighting both the greedy-choice and optimal substructure properties, using the stays ahead argument, prove the natural greedy algorithm given in class for the fractional knapsack problem always outputs an optimal solution.
- ✓ 2. ◊ Determine whether the two important characteristics of the Gale-Shapley's algorithm's output remain intact, when men propose and women accept/reject proposals based on their respective preference lists, while in every iteration a man for proposing is chosen uniformly at random from the set comprising free men who had not proposed to every woman. (Do prove those characteristics that would be present in the output of this modified algorithm.)
- ✓ 3. Given two ordered sequences $S = \{s_1, s_2, \dots, s_m\}$ and $T = \{t_1, t_2, \dots, t_n\}$, and a cost function $f : S \times T \rightarrow \mathbb{R}$, devise an efficient dynamic programming based algorithm to output a non-crossing alignment $\mathcal{A}$ between S and T that is of maximum cost with respect to f . Any non-crossing alignment $\mathcal{B}$ of S and T , (a) comprises a set of tuples (s_i, t_j) with $s_i \in S$ and $t_j \in T$, (b) for every two tuples $(s_i, t_j), (s_{i'}, t_{j'}) \in \mathcal{B}$ with $i < i'$, it is required $j < j'$, and (c) the cost of $\mathcal{B}$ is $\sum_{(s_i, t_j) \in \mathcal{B}} f(s_i, t_j)$.

$s_i \quad s_{i'}$
 $t_j \quad t_{j'}$
- 3 4. Given any set S of n points in the plane, devise a divide-and-conquer based algorithm to compute a simple polygon P with each of its vertices belonging to S such that for every triangle with its vertices belonging to S is contained in P . Your algorithm should take $O(n \lg n)$ time and $O(n)$ work-space in the worst-case. [To remind, a simple polygon is a polygon with interior of no edge intersects with another edge/vertex. Assume x -coordinates (resp. y -coordinates) of points in S are pairwise distinct. You may also assume no three points are collinear. No need to consider errors induced by floating-points; each word can hold a real number; assume operations supported by RAM model work for real numbers.]
5. In the local search algorithm for computing an approximate maximum cut of connected weighted simple undirected graphs, instead of moving any node v from one side to another side of the cut if that move improves the cut weight by 1, suppose any node v is moved only if the cut weight increases from W' to a value greater than $(1 + \frac{\epsilon}{n})W'$ for any W' . Prove that this modified strategy yields a weakly polynomial-time approximation algorithm. Here, $\epsilon > 0$ is an input parameter with $\frac{n}{\epsilon} > 1$. [You may require, $(1 + \frac{1}{x})^x \geq 2$ when $x \geq 1$.]